STAT 6560
Graphical Methods

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Project One

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CHARLES JOSEPH MINARD (1781-1870)

And The Best Statistical Graphic Ever Drawn

Citations: *`How others rate Minard's Flow Map of Napolean's Russian Campaign of 1812.`*

- "*the best statistical graphic ever drawn*" - (Tufte (1983), p. 40)
- Etienne-Jules Marey said "*it defies the pen of the historian in its brutal eloquence*" - (`http://en.wikipedia.org/wiki/Charles_Joseph_Minard`)
- Howard Wainer nominated it as the "*World's Champion Graph*" - (Wainer (1997) - `http://en.wikipedia.org/wiki/Charles_Joseph_Minard`)

Brief background

- Born on March 27, 1781.
- His father taught him to read and write at age 4.
- At age 6 he was taught a course on anatomy by a doctor.
- Minard was highly interested in engineering, and at age 16 entered a school of engineering to begin his studies.
- The first part of his career mostly consisted of teaching and working as a civil engineer. Gradually he became more research oriented and worked on private research thereafter.
- By the end of his life, Minard believed he had been the co-inventor of the flow map technique. He wrote he was pleased "at having given birth in my old age to a useful idea..." - (Robinson (1967), p. 104)

What was done before Minard?

Examples:

- **Late 1700's**: *Mathematical and chemical graphs begin to appear.*

- **William Playfair's 1801**: (*Chart of the National Debt of England*).
  - This line graph shows the increases and decreases of England's national debt from 1699 to 1800.
  - Demonstrates the beginning of line graphs.
- **Johann Heinrich Lambert**:
  - Lambert was one of few who used graphs extensively.
  - He used tables as graphs, as well as creating line and other graphs.
- **Louis-Ezechiel Pouchet's**: (*Pythagorean table*).
  - Pouchet helped France convert to the metric system using his graph of multiplications.
  - Often, we can use the multiplication table to help with calculations. However, decimal multiplications are more difficult than the multiplication table will allow. Pouchet created a graph to help with not only integer multiplication, but also decimal multiplication.
- **Gaspard Monge**:
  - Monge was one of the first to create a way to see 3-dimensional problems in 2-dimensions.
  - In his graph of the system of multiple projections, we see the process he went through in order to capture the problem on paper.

Minard's main graphical innovations

- Flow maps were his main innovation.
- He mapped population density using a numerical shading system.
- Minard used pie charts and maps to map zones of consumption and production.
- **Things to note about his graphs**:
  - His graphs are now called thematic maps, which are graphs that focus on the spatial characteristics of the distribution relative to geographical factors.

  - Minard would often change the geographical aspects in order to show the data how he intended. This practice is opposite of what we do today.
  - He always constructed his graphs such that the area representing the data was proportional to the data itself, whether it be pie charts or bar plots.

- **His Intuitive psychology**:
  - Color to compare categories.
  - Area to compare quantities.
  - Maps to show transportation of people and goods.

Some flaws in Minard's graphics

- Geological projections were often changed in order to show the data as he wanted it shown.
- He also revised coastlines and forced the geolocical scales to fit the data.
- Minard was quite aware of what he was doing, and called his maps *cartes figuratives.*

Links to Data file and R-code:

- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project1_minard_cities.txt`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project1_minard_troops.txt`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project1_minard_temps.txt`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project1_minard_napolean.R`
- `http://www.math.usu.edu/~symanzik/teaching/2009_stat6560/RDataAndScripts/anderson_jessica_project1_minard_canal.R`

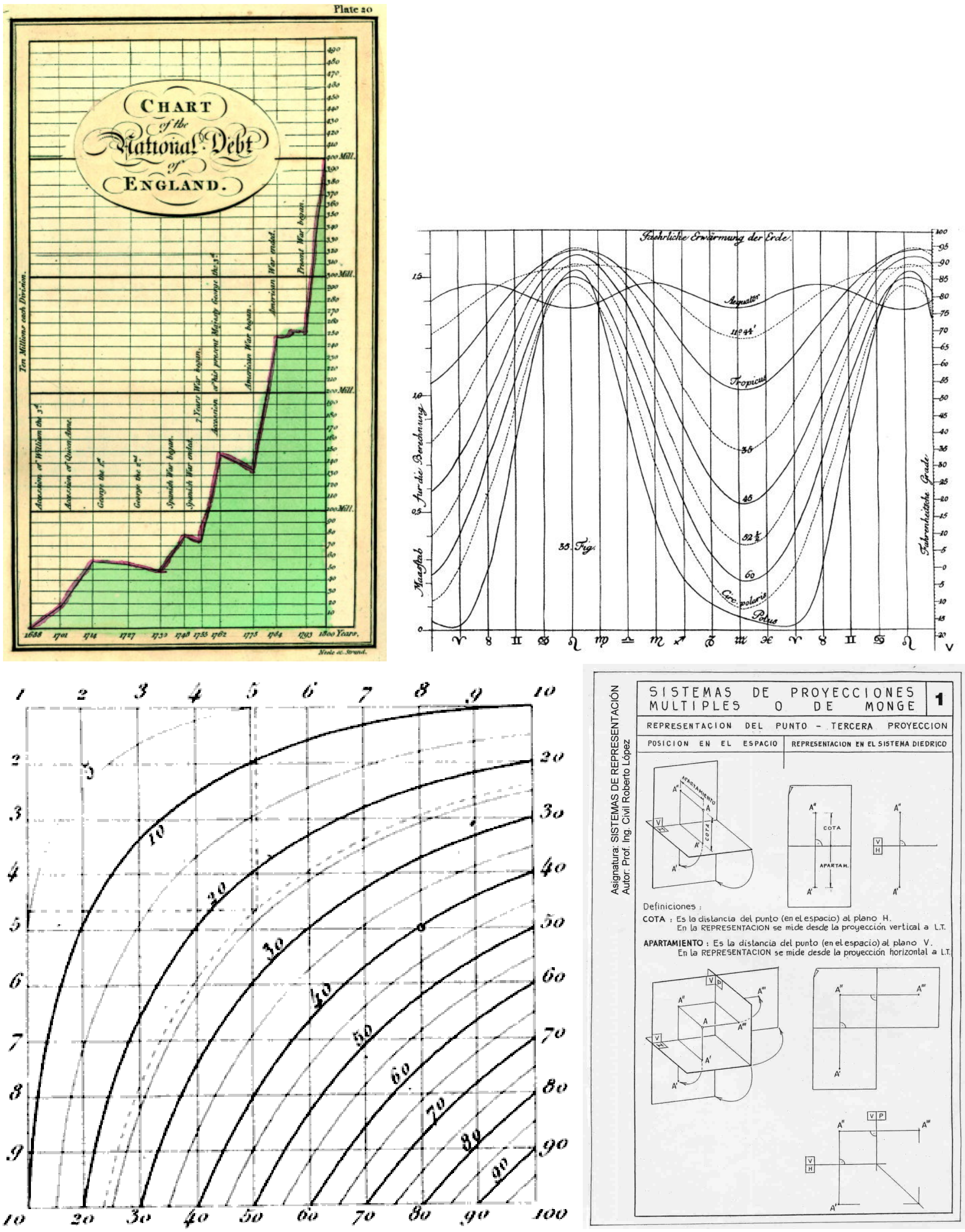

Figure 1: Playfair's chart of national debt (top left), Lambert's 1779 graph (top right), Pouchet's multiplication table (bottom left), and Monge's system of multiple projections (bottom right). *Source*: Figures taken from http://www.math.yorku.ca/SCS/Gallery/, http://www.math.yorku.ca/SCS/Gallery/images/lambert1779-graph.jpg, http://dekstop.de/weblog/2006/01/visualization_of_numeric_data/pouchet_multiplication.png and http://www.math.yorku.ca/SCS/Gallery/images/sist-monge.jpg.

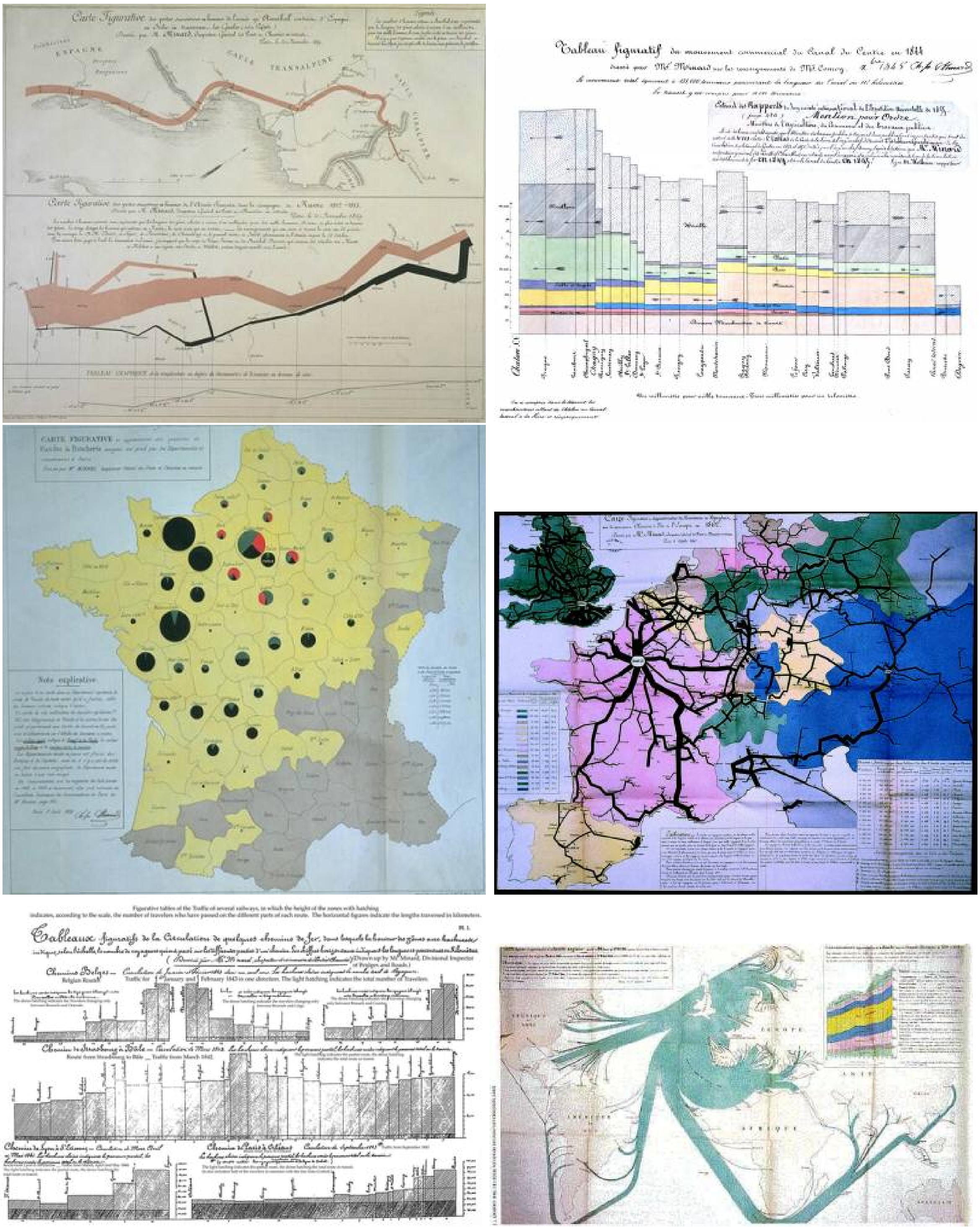

Figure 2: Selected Minard Graphs. Figures taken from `http://www.math.yorku.ca/SCS/Gallery/minbib/index.htm`

References

Robinson, A. H. (1967), 'The Thematic Maps of Charles Joseph Minard', *Imago Mundi* **21**, 95–108.

Tufte, E. R. (1983), *The Visual Display of Quantitative Information*, Graphics Press, Cheshire, CT.

Wainer, H. (1997), *Visual Revelations: Graphical Tales of Fate and Deception from Napoleon Bonaparte to Ross Perot*, Copernicus/Springer, New York, NY.